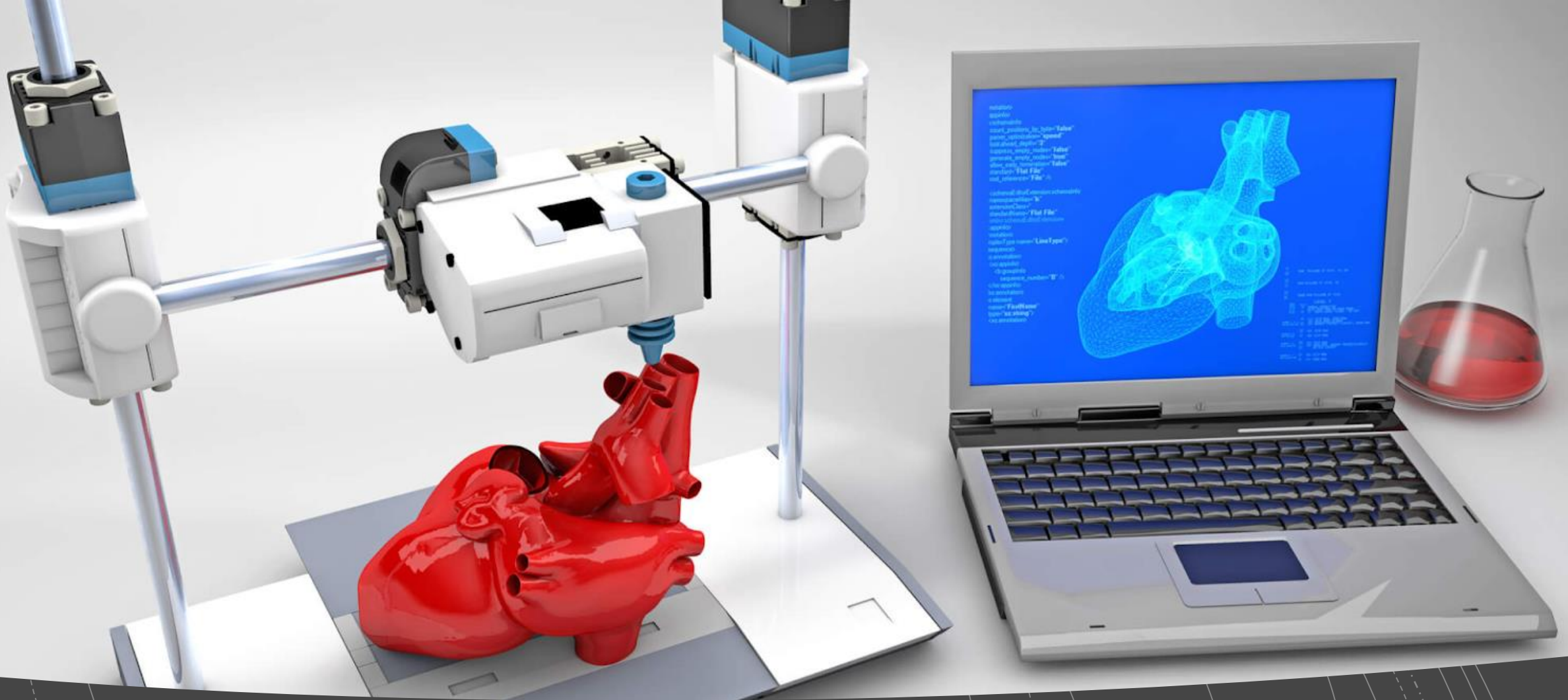
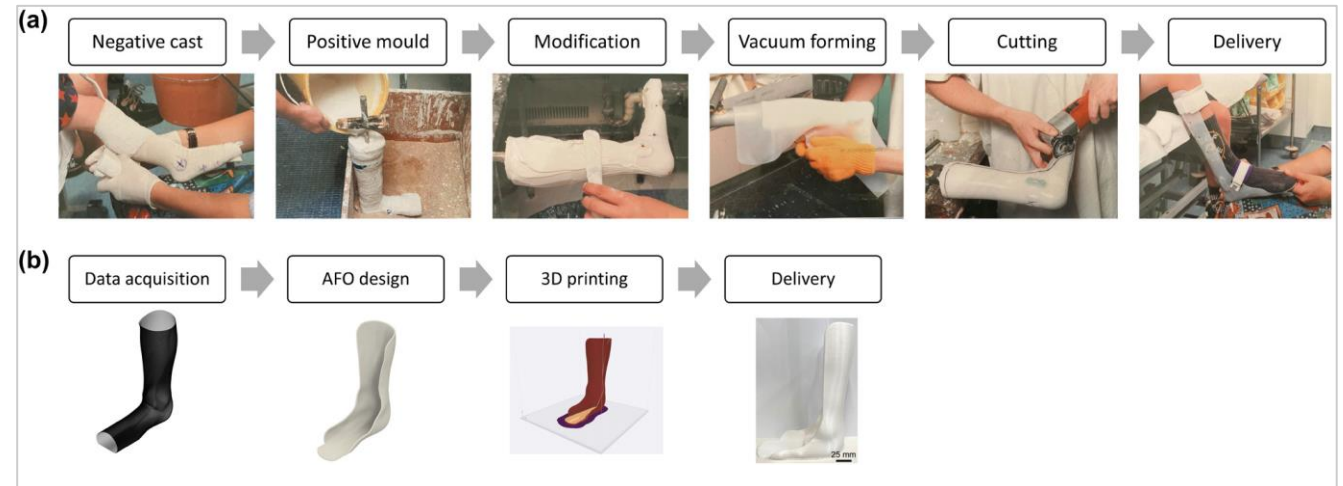




3D Printing in Medicine: Case Studies (2023–2025)

Exploring Real-World Applications and Outcomes
Presented by: Tushar Mitra, Hunter Tu

Case 1: Tuning Mechanical Responses of 3D-Printed AFOs (2024)



-Need: Conventional AFOs showed unpredictable stiffness due to print porosity and inconsistent infill densities.

-Technology: FDM printing using PLA and Nylon was optimized by altering infill percentage, shell thickness, and print orientation; mechanical testing measured tensile modulus and damping ratios.

-Outcome: Quantified 11–16% porosity and correlated stiffness reduction; enabled predictive tuning of gait-assist orthoses.

Source: Peng et al., 2024

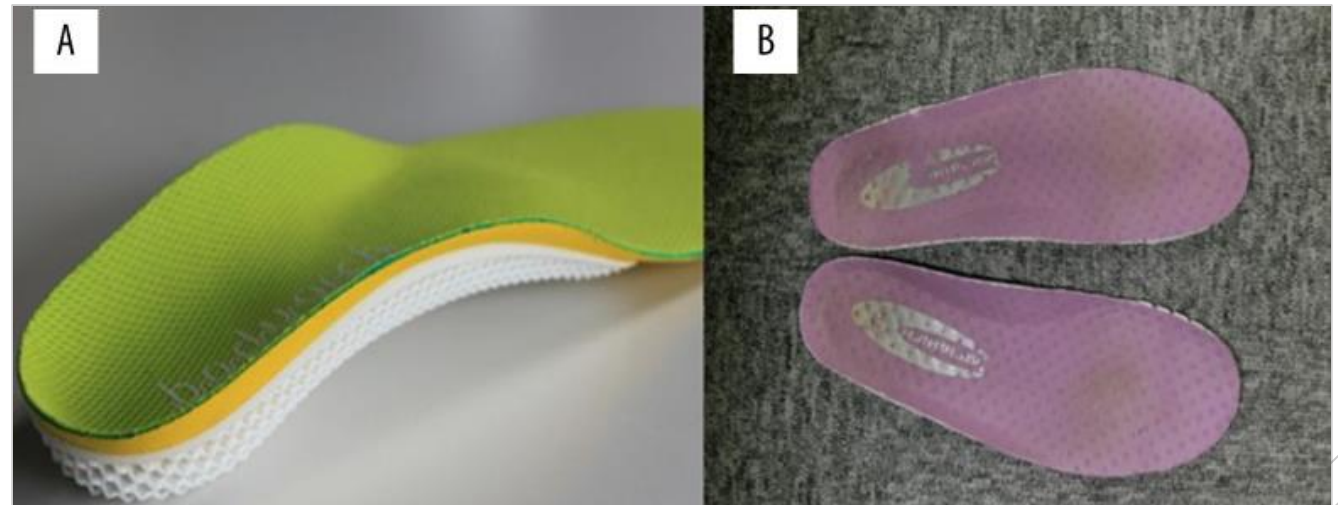
Case 2: Custom Insoles for Flatfoot Patients (2019)

-Need: Prefabricated insoles provided limited arch support and poor pressure redistribution.

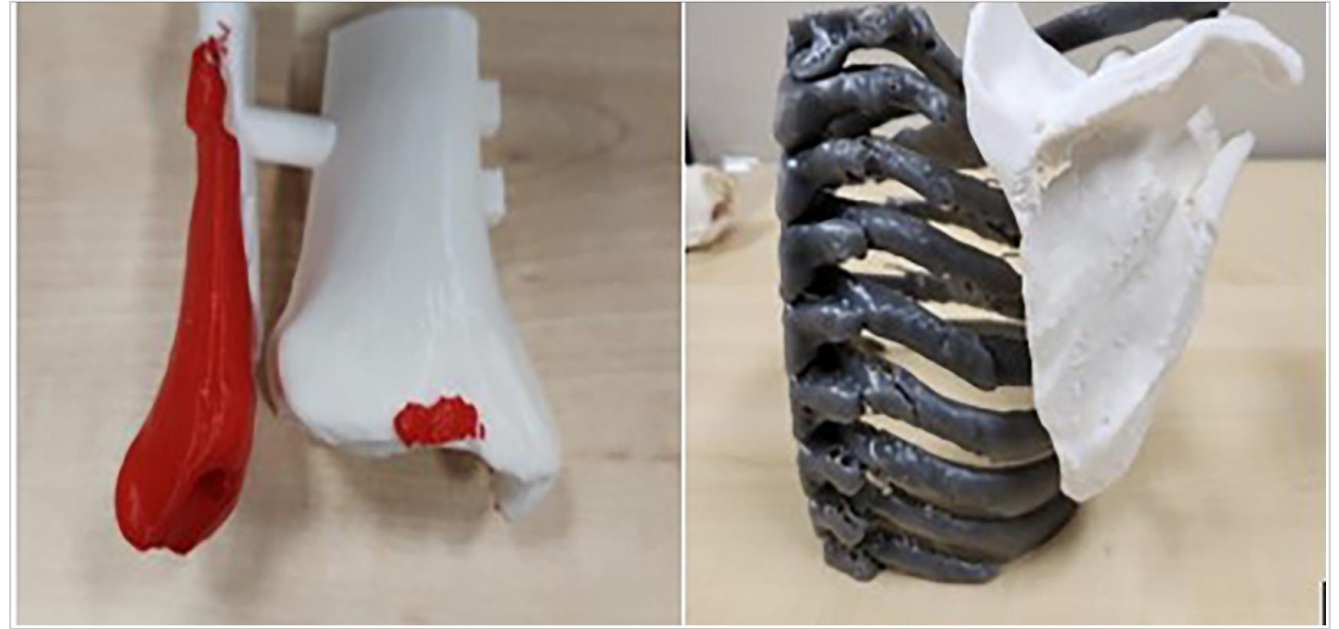
-Technology: 3D scanning of patient feet followed by polymer FDM printing with variable infill gradients in TPU to modulate elasticity.

-Outcome: Reduced peak plantar pressure by 15–20% and increased perceived comfort and adherence.

Source: Xu, 2019



Case 3: Anatomical Models for Doctor–Patient Communication (2024)



- Need:** Complex orthopedic conditions were hard to explain using 2D imaging.
- Technology:** 3D printed anatomical bone models using FDM and PETG polymers from CT data to display fractures and alignment corrections.
- Outcome:** Increased patient comprehension and satisfaction scores (4.8/5 vs 3.9/5 baseline) and reduced consultation time by 25%.

Source: Sanchez et al., 2024

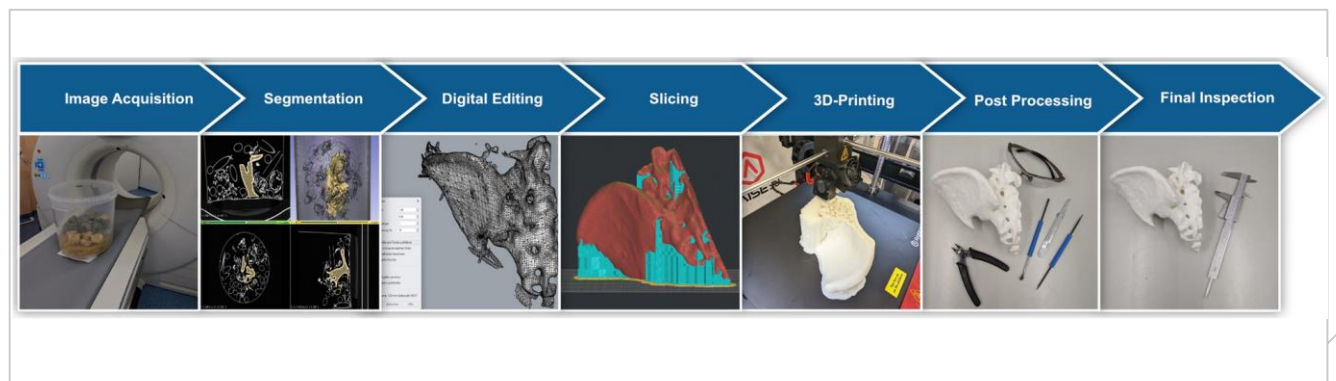
Case 4: Quality Assurance for Anatomical Models (2024)

-Need: Hospitals lacked standardized QA for accuracy in printed models.

-Technology: Multi-center comparison of segmentation, slicing, and printing workflows using CT-derived bone models printed in

-Outcome: Found dimensional deviations up to 2.3 mm and .825mm on average; proposed standard metrics for clinical model QA.

Source: Schhulze, 2024



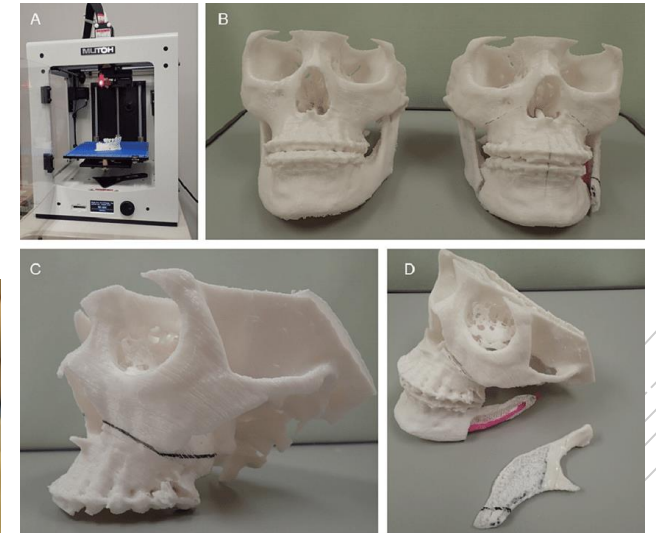
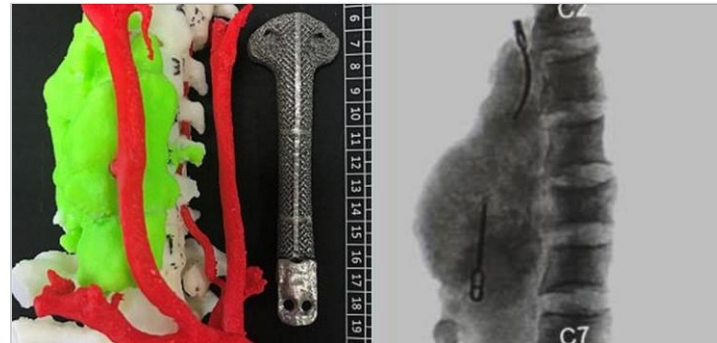
Case 5: Systematic Review on Planning & Training (2024)

-Need: Unclear impact of 3D printing on medical education and procedural outcomes.

-Technology: Meta-analysis of 40+ studies using PolyJet, SLA, and FDM for anatomical and surgical models.

-Outcome: 40% mean improvement in procedural performance and 27% reduction in operative time.

Source: Catasta et al., 2024



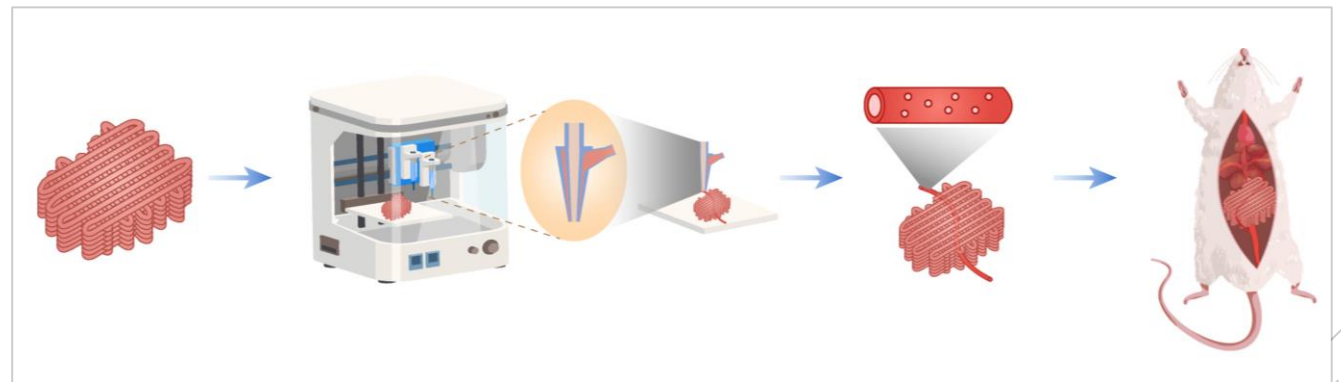
Case 6: Liver Bioprinting (2024)

-Need: Shortage of transplantable livers and ineffective in vitro models.

-Technology: Extrusion bioprinting using GelMA hydrogels and hepatocyte-laden bioinks to create vascularized liver constructs.

-Outcome: Improved albumin production and viability in mouse transplant models.

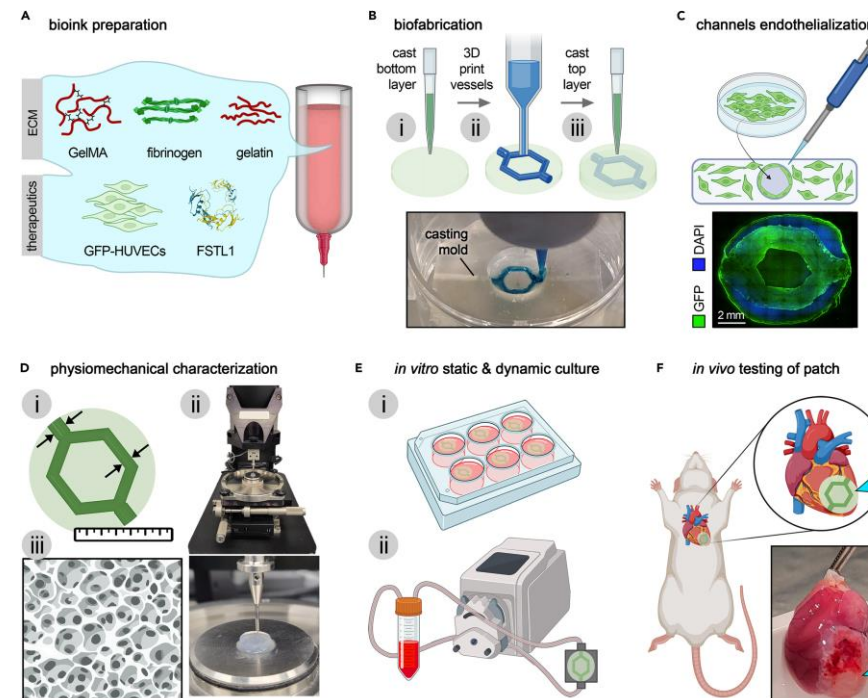
Source: Bo Deng et al., 2024



Case 7: Cardiac Patch Bioprinting (2025)

- Need:** Regeneration of infarcted myocardial tissue after heart attack.
- Technology:** Hydrogel-based extrusion bioprinting with endothelial and cardiomyocyte cells forming vascularized cardiac patches.
- Outcome:** Enhanced tissue integration and perfusion post-implantation in rodent models.

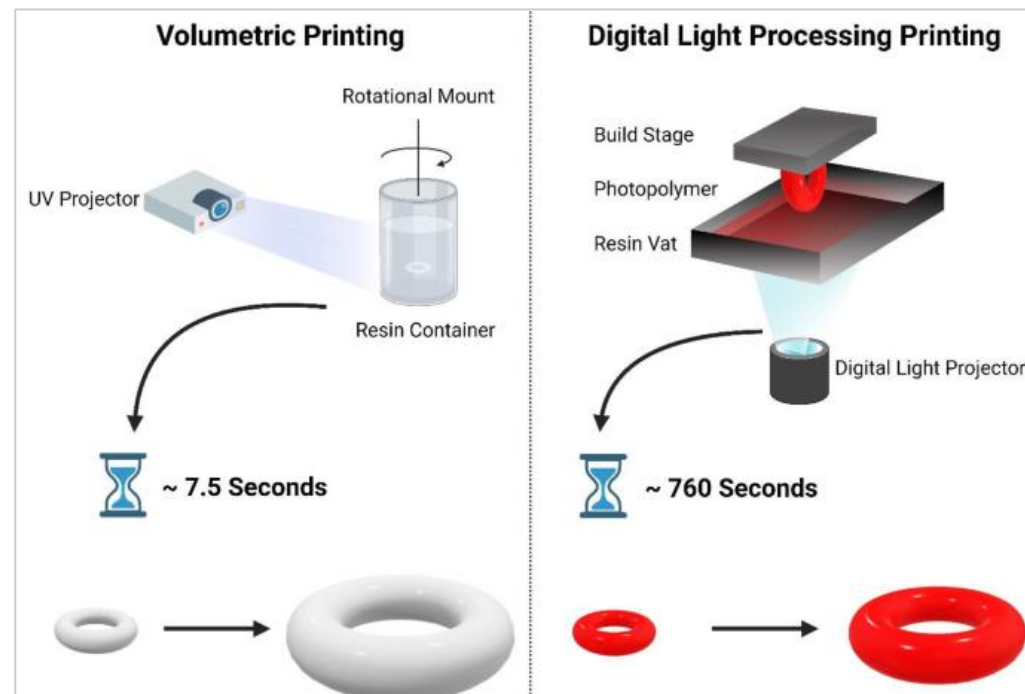
Source: Hwang, 2024



Case 8: Mutable Drug-Delivery Devices (2024)

- Need:** Rapid manufacturing of controlled-release pharmaceutical systems.
- Technology:** Volumetric photopolymerization of hydrogel matrices loaded with paracetamol; cured within 7.5 seconds.
- Outcome:** Tunable drug release kinetics and improved uniformity compared to conventional casting.

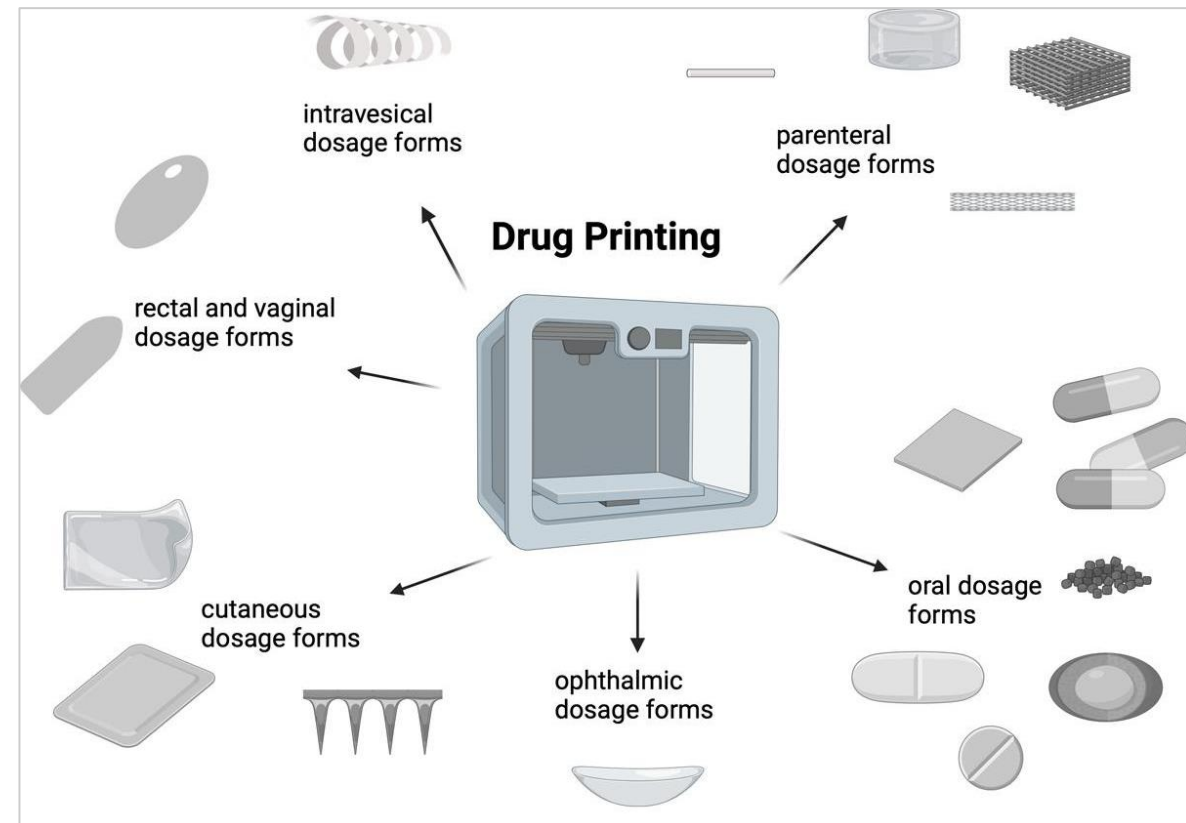
Source: Oh et al., 2025



Case 9: Dosage Form Printing (2024)

- Need:** Combine multiple drugs into single customizable tablets.
- Technology:** Binder-jet and FDM printing of layered pharmaceutical geometries with varied release profiles.
- Outcome:** Achieved independent control of dissolution rates and drug stability.

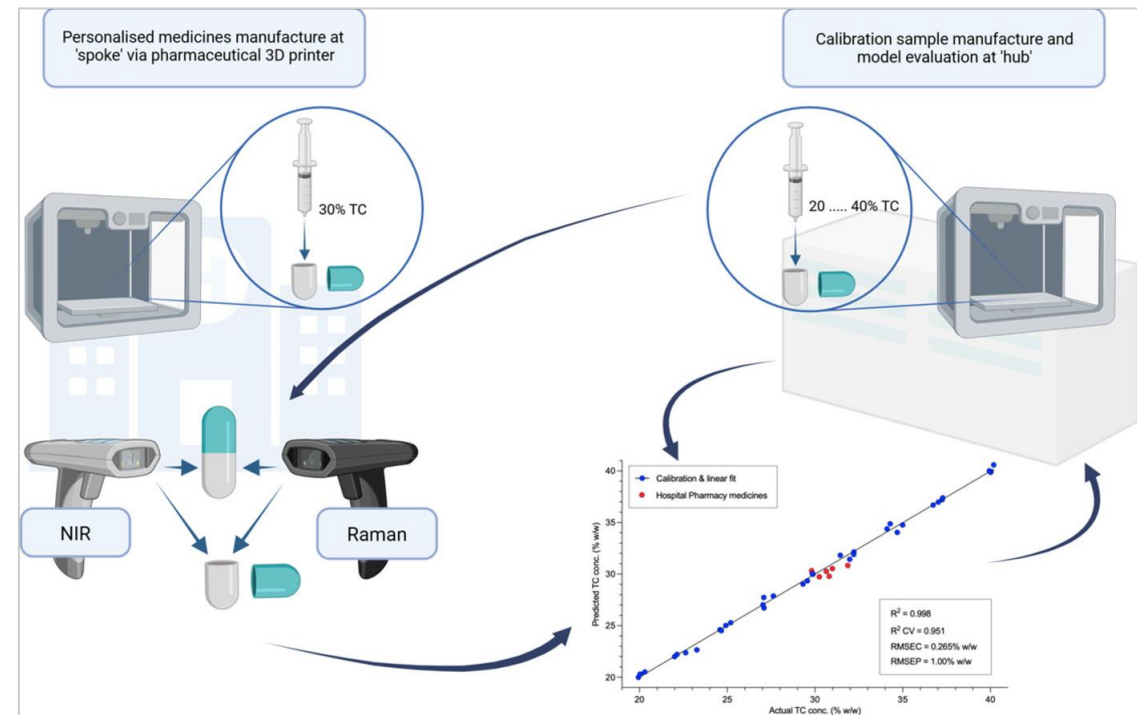
Source: Aeul et al., 2025



Case 10: Personalized Medications in Hospitals (2025)

- Need:** Personalized dosing for pediatric and geriatric patients in hospital settings.
- Technology:** Spectroscopically verified 3D-printed tablets manufactured on-demand using inkjet deposition.
- Outcome:** Demonstrated safe, accurate dosing and seamless clinical integration.

Source: Jørgensen et al., 2025



Summary & Future Directions

3D printing continues to revolutionize surgery, prosthetics, education, research, and drug delivery.

Future Directions:

- Validation and QA frameworks
- FDA regulatory adaptation
- Cost-benefit optimization
- Bioprinting scalability
- Algorithm-driven print parameterization

Q&A / Closing

Thank you to BMDC and the
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